Laboratory Record

of _____

Roll No.: _____
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AIM

Develop a JavaScript quiz app with a timer and scoring system using DOM manipulation.

Description

Developed a simple JavaScript Quiz App using DOM manipulation. The application displays multiple-choice questions, provides a 15-second timer for each question, allows users to select an option, automatically moves to the next question after selection or when time expires, and calculates the final score at the end of the quiz.

Algorithm

1. Setup: Initialize the HTML document and link the CSS and JavaScript files. Create variables for the quiz questions, current question, score, and timer.
2. Quiz Data: Store each question, its multiple-choice options, and the correct answer in a JavaScript array.
3. Question Display: Use DOM manipulation to display the current question and generate buttons for its options.
4. Timer: Start a 15-second countdown for each question and display the remaining time on the webpage.
5. Answer Selection: When the user selects an option, compare it with the correct answer and increase the score if the answer is correct.
6. Next Question: Automatically move to the next question when an option is selected or when the timer reaches zero.
7. Repeat: Continue displaying questions and resetting the timer until all questions are completed.
8. Score Calculation: After the last question, stop the timer and calculate the final score.
9. Result Display: Use DOM manipulation to display the final score to the user.

Program Code

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Quiz App</title>
  <link rel="stylesheet" href="style.css">
</head>
```

```
<body>
  <div class="quiz-box">
    <h1>Simple Quiz</h1>
    <div class="timer">
      Time Left: <span id="timer">15</span>s
    </div>
    <h2 id="question">Question</h2>
    <div id="options"></div>
    <button id="next-btn">Next</button>
    <p id="score"></p>
    <p id="result"></p>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

Script.js

```
const quiz = [
  {
    question: "Which language runs in a web browser?",
    options: ["Python", "Java", "JavaScript", "C++"],
    answer: "JavaScript"
  },
  {
    question: "What does DOM stand for?",
    options: [
      "Document Object Model",
      "Data Object Management",
      "Digital Ordinance Model",
```

```
            "Desktop Oriented Mode"
        ],
        answer: "Document Object Model"
    },
    {
        question: "Which keyword declares a constant?",
        options: ["var", "let", "const", "static"],
        answer: "const"
    },
    {
        question: "Which HTML tag links JavaScript?",
        options: ["<style>", "<script>", "<link>", "<js>"],
        answer: "<script>"
    }
];
let index = 0;
let score = 0;
let time = 15;
let timer;
const question = document.getElementById("question");
const options = document.getElementById("options");
const timerEl = document.getElementById("timer");
const next = document.getElementById("next-btn");
const scoreEl = document.getElementById("score");
function loadQuiz() {
    clearInterval(timer);
    time = 15;
    timerEl.textContent = time;
```

```
  question.textContent = quiz[index].question;
  options.innerHTML = "";
  quiz[index].options.forEach(option => {
    const btn = document.createElement("button");
    btn.textContent = option;
    btn.onclick = () => {
      if (option === quiz[index].answer) score++;
      nextQuestion();
    };
    options.appendChild(btn);
  });
  timer = setInterval(() => {
    time--;
    timerEl.textContent = time;
    if (time === 0) {
      nextQuestion();
    }
  }, 1000);
}
function nextQuestion() {
  clearInterval(timer);
  index++;
  if (index < quiz.length) {
    loadQuiz();
  } else {
    question.textContent = "Quiz Completed!";
    options.innerHTML = "";
    next.style.display = "none";
```

```
        timerEl.textContent = "0";
        scoreEl.textContent = `Score: ${score}/${quiz.length}`;
    }
}
next.onclick = nextQuestion;
loadQuiz();
```

Style.css

```
body {
    font-family: Arial, sans-serif;
    background: #f5f5f5;
}
.quiz-box {
    width: 350px;
    margin: 100px auto;
    padding: 20px;
    background: white;
    border: 1px solid #ddd;
    border-radius: 8px;
}
h1 {
    text-align: center;
    font-size: 24px;
}
.timer {
    text-align: center;
    margin: 15px 0;
}
#question {
    font-size: 18px;
```

```
}
#options button {
    display: block;
    width: 100%;
    padding: 10px;
    margin: 8px 0;
    background: white;
    border: 1px solid #ccc;
    border-radius: 5px;
    text-align: left;
}
#next-btn {
    display: block;
    margin: 15px auto;
    padding: 8px 20px;
    border: none;
    background: #333;
    color: white;
    border-radius: 5px;
}
#next-btn:disabled {
    background: #ccc;
}
#score,
#result {
    text-align: center;
}
#options button:hover {
    background: #eee;
```

```
}
#next-btn:hover {
   background: #555;
}
```









